

MCI 102 - Mathematics-II

Differential Equations

Find a basis if one solution is known: Method of reduction of order.

Let us consider the second order linear homogeneous ODE

$$y'' + p(x)y' + q(x)y = 0. \quad (1)$$

As remarked earlier, there is no general method to find a basis of solutions of (1). It happens quite often that one non zero solution can be found by inspection or in some other way.

Then a second LI solution can be obtained by taking $y_2 = v(x)y_1$, where y_1 is a first non zero solution & $v(x)$ is a non-constant function. We illustrate this method by an example.

Example 1: If $y_1 = x$ is a solution of $(1-x^2)y'' - 2xy' + 2y = 0$, find a basis of solutions and a general solution.

Solution: Let $y_2 = v y_1 = vx$ be a second LI solution.

Differentiate w.r. to x , $y_2' = v + v'x$.

Again differentiate w.r. to x , $y_2'' = v' + v' + v''x = 2v' + v''x$.

If y_2 is a solution of given ODE, it satisfies the given ODE.

$$\therefore (1-x^2)y_2'' - 2xy_2' + 2y_2 = 0.$$

$$\Rightarrow (1-x^2)(2v' + v''x) - 2x(v + v'x) + 2vx = 0$$

$$\Rightarrow (1-x^2)xv'' + 2(1-x^2)v' - 2x^2v' - 2xv + 2xv = 0$$

$$\Rightarrow (1-x^2)xv'' + 2(1-2x^2)v' = 0.$$

Take $v' = u$, this gives $(1-x^2)xu' + 2(1-2x^2)u = 0$.

$$\Rightarrow \frac{du}{u} + \frac{2(1-2x^2)}{x(1-x^2)} dx = 0$$

$$\Rightarrow \frac{du}{u} = - \left(\frac{2}{x} - \frac{1}{1-x} + \frac{1}{1+x} \right) dx = 0.$$

$$\begin{aligned} \text{On integrating, } \ln u &= -(2 \ln x + \ln(1-x) + \ln(1+x)) \\ \Rightarrow \ln u &= -\ln x^2(1-x^2) \quad (\text{Ignoring the constant of integration}) \\ \Rightarrow u &= \frac{1}{x^2(1-x^2)} \end{aligned}$$

$$\text{Thus, } v' = u = \frac{1}{x^2(1-x^2)} \Rightarrow dv = \left(\frac{1}{x^2} + \frac{1}{1-x^2} \right) dx$$

$$\Rightarrow dv = \left(\frac{1}{x^2} + \frac{1}{2(1-x)} + \frac{1}{2(1+x)} \right) dx.$$

Again by integrating and ignoring the constant of integration,

$$v = -\frac{1}{2} - \frac{1}{2} \ln(1-x) + \frac{1}{2} \ln(1+x).$$

$$\therefore y_2 = x \left(-\frac{1}{2} + \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) \right) = -1 + \frac{1}{2} x \ln \left(\frac{1+x}{1-x} \right).$$

Here $\{y_1, y_2\}$ is a basis and a general solution is given by
 $y = C_1 x + C_2 \left(-1 + \frac{1}{2} x \ln \left(\frac{1+x}{1-x} \right) \right).$ $\square$

Exercise 1: If y_1 is a non-zero solution of Eq. (1) and $y_2 = v y_1$ is a second LI solution of Eq. (1), then show that

$$v = \int \frac{\bar{c} \int p dx}{y_1^2} dx.$$

Hint: Follow the steps of the above example. LI follows from Wronskian test. (Here $W = e^{-\int p dx} \neq 0$).

Exercise 2: If $y_1 = \frac{\cos x}{x}$ is a solution of $xy'' + 2y' + xy = 0$, find a general solution. (Here, $p = 2/x$ & $q = 1$).

Homogeneous Linear ODEs with constant coefficients:

We now consider second order homogeneous linear ODE

$$y'' + ay' + by = 0, \quad (2)$$

where a and b are constants.

We recall that the solution of the first order linear ODE

$y' + ky = 0$, k is a constant, is an exponential function

$y = c e^{-kx}$. This gives us the idea to try this as a solution of Eq. (2). Suppose $y = e^{mx}$ is a solution of Eq. (2).

This gives, $y' = m e^{mx}$, $y'' = m^2 e^{mx}$ and
 $m^2 e^{mx} + a m e^{mx} + b e^{mx} = 0.$

Since $e^{mx} \neq 0$, $m^2 + am + b = 0.$ (3)

The above eq. is called the characteristic eq. Hence, if

m is a root of Eq. (3), then $y = e^{mx}$ is a solution of Eq. (2).

We have three cases for solutions of Eq. (2) depending upon roots of (3).

Case I: If m_1 and m_2 are real roots and $m_1 \neq m_2$, then $y_1 = e^{m_1 x}$ and $y_2 = e^{m_2 x}$ are LI solutions of Eq. (2). So the general solution is $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}.$

Case II: If m_1 and m_2 are real and $m_1 = m_2$, then $y_1 = e^{m_1 x}$ and $y_2 = x e^{m_1 x}$ (use the formula given in Exercise 1 to find

y_2) are LI solutions of Eq(2). So the general solution is

$$y = c_1 e^{m_1 x} + c_2 x e^{m_1 x}.$$

Case III: Suppose m_1 and m_2 are complex roots. If $m_1 = \alpha + i\beta$, then $m_2 = \alpha - i\beta$. Thus $e^{m_1 x} = e^{(\alpha + i\beta)x} = e^{\alpha x} (\cos \beta x + i \sin \beta x)$ and $e^{m_2 x} = e^{(\alpha - i\beta)x} = e^{\alpha x} (\cos \beta x - i \sin \beta x)$. As $m_1 \neq m_2$, $e^{m_1 x}$ & $e^{m_2 x}$ are LI solutions of Eq(2), the general solution is

$$y = c_1 e^{\alpha x} \cos \beta x + c_2 e^{\alpha x} \sin \beta x.$$

Exercise: In Case III, show that $y_1 = e^{\alpha x} \cos \beta x$ & $y_2 = e^{\alpha x} \sin \beta x$ are LI solutions of Eq(2)

Example 2: Find the general solution of the following ODEs
ci) $y'' - 6y' - 7y = 0$ cii) $y'' + 4y' + 4y = 0$ ciii) $y'' + 9y = 0$.

Solution: ci) Put $y = e^{mx}$ in $y'' - 6y' - 7y = 0$, this gives,

$$m^2 - 6m - 7 = 0 \Rightarrow (m - 7)(m + 1) = 0 \\ \Rightarrow m = 7, -1.$$

By Case I, the general solution is
 $y = c_1 e^{7x} + c_2 e^{-x}.$

(ii) Put $y = e^{mx}$ in $y'' + 4y' + 4y = 0$, this gives

$$m^2 + 4m + 4 = 0 \Rightarrow (m + 2)^2 = 0 \\ \Rightarrow m = 2, 2.$$

Thus, the general solution is $y = c_1 e^{2x} + c_2 x e^{2x}$. (Case II)

(iii) Put $y = e^{mx}$ in $y'' + 9y = 0$, this gives
 $m^2 + 9 = 0 \Rightarrow m = 3i, -3i.$

Thus, the general solution is $y = c_1 \cos 3x + c_2 \sin 3x$. (Case III)

Higher order homogeneous linear ODEs:

For simplicity, we fix the following notations:

$$\frac{d}{dx} = D, \quad \frac{d^2}{dx^2} = D^2, \quad \dots, \quad \frac{d^n}{dx^n} = D^n.$$

The standard form of n^{th} order linear ODEs is given by

$$D^n y + p_{n-1}(x) D^{n-1} y + \dots + p_1(x) D y + p_0(x) y = r(x), \quad (4)$$

the coefficients $p_0, p_1, \dots, p_{n-1}$ and r are functions of x .
If $r(x) = 0$, then the ODE is called homogeneous ODE, otherwise it is called non-homogeneous.

Theorem: If $y_1, y_2, \dots, y_n$ are solutions of homogeneous linear ODE $(D^n + p_{n-1}(x)D^{n-1} + \dots + p_1(x)D + p_0(x))y = 0$ (5) on open interval I , then $y = c_1 y_1 + c_2 y_2 + \dots + c_n y_n$, c_i 's are constant, is also a solution of (5) on I .

Definition: The functions $y_1, y_2, \dots, y_n$ are called LI on some open interval I if $c_1 y_1(x) + \dots + c_n y_n(x) = 0$ for all x in I and constant c_i 's then $c_1 = c_2 = \dots = c_n = 0$.

Definition (General solution): The general solution of Eq. (5) on an open interval I is a solution of the form $y = c_1 y_1 + c_2 y_2 + \dots + c_n y_n$, where c_i 's are constant and $y_1, y_2, \dots, y_n$ are LI solutions of Eq. (5) on I .

Wronskian: Let $y_1, y_2, \dots, y_n$ be n -times differentiable functions. We define the Wronskian of $y_1, y_2, \dots, y_n$ as

$$W(x) = \det \begin{bmatrix} y_1 & y_2 & \dots & y_n \\ y_1' & y_2' & \dots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \dots & y_n^{(n-1)} \end{bmatrix}.$$

Exercise: Write the test for LI of $y_1, y_2, \dots, y_n$ using the Wronskian as in the previous lecture note and give the proof.

Exercise: If $m_1, m_2, \dots, m_n$ are all distinct, show that $e^{m_1 x}, e^{m_2 x}, \dots, e^{m_n x}$ are LI.

Exercise: Show that $e^{mx}, xe^{mx}, x^2 e^{mx}, \dots, x^{n-1} e^{mx}$ are LI.

Theorem: If the coefficients $p_0, p_1, \dots, p_{n-1}$ of Eq. (5) are continuous on some open interval I then Eq. (5) has a general solution on I .

Hint: Generalize the proof of second order to n th-order.

Exercise: If ODE Eq.(5) has continuous coefficients on some open interval I , then every solution on I is of the form

$$y = c_1 y_1 + \dots + c_n y_n,$$

where $y_1, y_2, \dots, y_n$ are LI solutions of (5) on I .

Homogeneous linear ODE with constant coefficients (Higher order):

The general form of higher order homogeneous linear ODE with constant coefficients is given by

$$(D^n + a_{n-1}D^{n-1} + \dots + a_1D + a_0)y = 0, \quad (6)$$

where a_i 's are constant.

As in the second order ODE case, we put $y = e^{mx}$ in Eq.(6) to obtain the characteristic equation

$$m^n + a_{n-1}m^{n-1} + \dots + a_1m + a_0 = 0. \quad (\because D^i e^{mx} = m^i e^{mx})$$

Case I: If $m_1, m_2, \dots, m_n$ are all distinct and real roots, then the general solution of Eq.(6) is given by

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \dots + c_n e^{m_n x}$$

Case II: If $m_1 = m_2 = \dots = m_k$ and $m_{k+1}, m_{k+2}, \dots, m_n$ are distinct real roots, the general solution is given by

$$y = c_1 e^{m_1 x} + c_2 x e^{m_1 x} + \dots + c_k x^{k-1} e^{m_1 x} + c_{k+1} e^{m_{k+1} x} + \dots + c_n e^{m_n x}.$$

Case III: If some of the roots are complex (say $m_i = \alpha + i\beta$), then we can use $e^{(\alpha+i\beta)x} = e^{\alpha x} (\cos \beta x + i \sin \beta x)$ and $e^{(\alpha-i\beta)x} = e^{\alpha x} (\cos \beta x - i \sin \beta x)$ to get the real form of the general solution (like in the second order ODE).

Example 3: Find the general solution of the following ODEs.

(i) $(D^3 - 2D^2 - D + 2)y = 0$

(ii) $(D^3 + D^2 - D - 1)y = 0.$

(iii) $(D^3 - 3D^2 + 3D - 1)y = 0$

(iv) $(D^3 - 1)y = 0.$

(v) $(D^4 - 1)y = 0.$

Solution: (i) Put $y = e^{mx}$ in $(D^3 - 2D^2 - D + 2)y = 0$. This gives

$$m^3 - 2m^2 - m + 2 = 0 \Rightarrow m^2(m-2) - (m-2) = 0$$

$$\Rightarrow (m^2 - 1)(m - 2) = 0 \Rightarrow m = 1, -1, 2.$$

Thus, the general solution is

$$y = c_1 e^x + c_2 e^{-x} + c_3 e^{2x}. \quad (\text{Case I})$$

cii) Put $y = e^{mx}$ in $(D^3 + D^2 - D - 1)y = 0$ to get
 $m^3 + m^2 - m - 1 = 0 \Rightarrow m^2(m+1) - (m+1) = 0$.
 $\Rightarrow (m^2 - 1)(m+1) = 0 \Rightarrow m = 1, -1, -1$.

Thus, the general solution is
 $y = c_1 e^x + c_2 e^{-x} + c_3 x e^{-x}$. (Case II)

ciii) Put $y = e^{mx}$ in $(D^3 - 3D^2 + 3D - 1)y = 0$ to get
 $m^3 - 3m^2 + 3m - 1 = 0 \Rightarrow (m-1)^3 = 0$
 $\Rightarrow m = 1, 1, 1$.

Thus, the general solution is
 $y = c_1 e^x + c_2 x e^x + c_3 x^2 e^x$. (Case II)

civ) Put $y = e^{mx}$ in $(D^3 - 1)y = 0$ to get
 $m^3 - 1 = 0 \Rightarrow (m-1)(m^2 + m + 1) = 0$
 $m = 1, \frac{-1+i\sqrt{3}}{2}, \frac{-1-i\sqrt{3}}{2}$.

Thus, the general solution is
 $y = c_1 e^x + c_2 e^{-x/2} \cos \frac{\sqrt{3}}{2} x + c_3 e^{-x/2} \sin \frac{\sqrt{3}}{2} x$. (Case III)

v) Put $y = e^{mx}$ in $(D^4 - 1)y = 0$, this gives
 $m^4 - 1 = 0 \Rightarrow (m^2 - 1)(m^2 + 1) = 0$
 $\Rightarrow m = 1, -1, i, -i$.

Thus, the general solution is
 $y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x$. (Case III)

Remark: The general solution of homogeneous linear ODE (Eqs (5) & (6)) is also called the complementary function and in short hand notation it is denoted by C.F. $\square$

Exercise: Show that $y_1 = e^{ax}$ is a solution of
 $(D^3 - 3aD^2 + 3a^2D - a^3)y = 0$. — (*)

If $y = v y_1$ is another solution of (*), then show that
 $v = c_1 + c_2 x + c_3 x^2$.

Exercise: Generalize the above exercise to n-order diff. eq.

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